PROBLEM CL3

Double-Elbow Mouthpiece: Exit Velocity, Force and Moment

Original statement · CL3.pdf

Control volume | Linear momentum | Angular momentum | Gauge pressure

OFFICIAL PROBLEM STATEMENT

The double-elbow mouth piece of exit cross section A_o shown in the figure of CL3.pdf is to be installed at the end of a circular pipe of cross section A_p to redirect the outflow of water into the open atmosphere. Once the mouth piece is installed, the uniform values U_p and p_p are measured for the flow velocity and the pressure at the end of the pipe (i.e., just at the entrance of the mouth piece). Find the value of the uniform velocity at the exit section, U_o, as well as the force exerted on the mouth piece and its corresponding moment with respect to the entrance section. (Handwritten addition: neglect the effect of gravity.)

Source: CL3.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eqs. 3.37-3.38, force and moment on a body Eqs. 3.32-3.33.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible (water); neglect $\bar{g}$ (handwritten addition); uniform properties at inlet and outlet; viscous stresses negligible on both sections.
- Free jet at the exit: $p = p_a$ over Σ_o. The thin wall of the mouth piece has the atmosphere p_a on its outer face; the force asked for is the net force of the water inside plus the atmosphere outside, as in the official solution. Ignoring the atmosphere would add $p_a(A_p - A_o)\,\bar{e}_x$.
- Geometry (figure): S-shaped piece; the exit axis is a vertical distance H below the entrance axis; both the entrance and the exit flows are along $+\bar{e}_x$.
- Data: $[\rho,\ A_p,\ A_o,\ U_p,\ p_p,\ p_a,\ H]$. Unknowns: $[U_o,\ \bar{F}_{MP},\ \bar{M}_{MP}]$.

Symbol	Meaning	SI unit
A_p, A_o	entrance and exit areas	m^2
U_p, p_p	velocity and pressure at the entrance	m/s, Pa
U_o	exit velocity	m/s
H	vertical offset between the axes	m
$\bar{F}_{MP}$, $\bar{M}_{MP}$	force and moment on the mouth piece	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Inertial frame fixed to the pipe: $\bar{e}_x$ along the pipe axis (flow direction), $\bar{e}_y$ vertical upwards, $\bar{e}_z = \bar{e}_x \wedge \bar{e}_y$ towards the reader. Origin O at the centre of the entrance section (moments about O).
- Centre of the exit section: $\bar{x}_o = X\bar{e}_x - H\bar{e}_y$ relative to O (X drops out). Position within a circular section relative to its centre: $\bar{\eta}$, with $\int \bar{\eta}\, d\sigma = 0$.
- Control volume: the water inside the mouth piece, fixed ($\bar{v}_c = 0$). Surfaces: Σ_p (entrance, $\bar{n} = -\bar{e}_x$), Σ_o (exit, $\bar{n} = +\bar{e}_x$), Σ_w (inner wall).

Force and moment of the water and the atmosphere on the mouth piece, with $\bar{n}$ the outward normal of the control volume:

$$\boxed{\bar{F}_{MP} = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

$$\boxed{\bar{M}_{MP} = \int_{\Sigma_w} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_p	$\rho\,U_p\bar{e}_x\cdot(-\bar{e}_x)\,A_p = -\rho U_p A_p$
Σ_o	$\rho\,U_o\bar{e}_x\cdot\bar{e}_x\,A_o = +\rho U_o A_o$
Σ_w	0 (wall)

$$\rho U_p A_p = \rho U_o A_o \implies \boxed{U_o = \frac{A_p}{A_o}\,U_p}, \qquad \dot{m} \equiv \rho U_p A_p = \rho U_o A_o \quad (1)$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_p	$\rho U_p\bar{e}_x\,(-U_p)\,A_p = -\dot{m}\,U_p\bar{e}_x$	$-(p_p-p_a)(-\bar{e}_x)A_p = (p_p-p_a)A_p\bar{e}_x$	0 (uniform)
Σ_o	$\rho U_o\bar{e}_x\,(U_o)\,A_o = \dot{m}\,U_o\bar{e}_x$	0 (free jet, $p=p_a$)	0 (uniform)

Surface	$\int \rho \bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}_{MP}$.

$$\dot{m}\,(U_o-U_p)\,\bar{e}_x=(p_p-p_a)A_p\,\bar{e}_x-\bar{F}_{MP}\implies\boxed{\bar{F}_{MP}=\left[(p_p-p_a)A_p+\rho U_p\,(U_p-U_o)\,A_p\right]\bar{e}_x}\quad(2)$$

No entry of the table has a component along $\bar{e}_y$ or $\bar{e}_z$, so $F_{MP,y}=F_{MP,z}=0$.

Angular momentum about O (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\,\bar{x}\wedge\bar{v}\,dV}_{0}+\int_{\Sigma_c}\rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma=-\int_{\Sigma_c}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma+\underbrace{\int_{V_c}\rho\,\bar{x}\wedge\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_p	$-\rho U_p^2\left(\int\bar{\eta}\,d\sigma\right)\wedge\bar{e}_x=0$	$(p_p-p_a)\left(\int\bar{\eta}\,d\sigma\right)\wedge\bar{e}_x=0$	0
Σ_o	$\rho U_o^2 A_o\,(\bar{x}_o\wedge\bar{e}_x)=\rho U_o^2 H A_o\,\bar{e}_z$	$0\;(p=p_a)$	0
Σ_w	0	$-\int_{\Sigma_w}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed Σ_w terms: $-\bar{M}_{MP}$. In the Σ_o entry, $\bar{x}=\bar{x}_o+\bar{\eta}$ and $\bar{x}_o\wedge\bar{e}_x=(X\bar{e}_x-H\bar{e}_y)\wedge\bar{e}_x=-H\,\bar{e}_y\wedge\bar{e}_x=+H\,\bar{e}_z$.

$$\rho U_o^2 H A_o\,\bar{e}_z=-\bar{M}_{MP}\implies\boxed{\bar{M}_{MP}=-\rho U_o^2 H A_o\,\bar{e}_z=-\rho\,\frac{A_p^2}{A_o}\,U_p^2\,H\,\bar{e}_z}\quad(3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $A_o < A_p$ gives $U_o > U_p$: the piece accelerates the water, and the wall supplies the momentum change.
- The moment exists only because the exit axis is offset by H: for $H \to 0$, $\bar{M}_{MP} \to 0$. With the exit below the entrance, the moment is clockwise as seen in the figure ($\bar{e}_z$ towards the reader).
- Limit $A_o \to A_p$: $U_o = U_p$ and $\bar{F}_{MP} \to (p_p - p_a) A_p \bar{e}_x$, the pure pressure force.
- Dimensions (SI): $[\rho U_p A_p (U_p - U_o)] = \mathrm{N}$; $[\rho U_o^2 H A_o] = \mathrm{N \cdot m}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $A_p = 0.02$ m^2, $A_o = 0.008$ m^2, $U_p = 3$ m/s, $p_p = 2.5 \times 10^5$ Pa, $p_a = 101325$ Pa, $H = 1.5$ m give $U_o = 7.5$ m/s, $F_{MP,x} = 2703.5$ N and $M_{MP,z} = -675$ N·m.
- Agreement with the official solution: CL3.pdf obtains (1), (2) and (3).

Extension (beyond the official solution)

If the flow inside were lossless (not stated), Bernoulli's equation $p_p - p_a = \frac{1}{2}\rho(U_o^2 - U_p^2)$ would give $F_{MP,x} = \frac{1}{2}\rho A_p (U_o - U_p)^2 \geq 0$. With losses, the measured p_p must be used, as in (2).